Statement E.61 (Proved). The exact mod-twelve genus-one sewing projection. *Use the positive generators of E.1 and E.58. Let $(1, f)$ be any normalized full gentle pair, with its actual extension*

$$\Phi(s) = sf = \beta s \beta^{-1}, \qquad \Phi(r) = f^{-1} r, \qquad \Phi(sr) = sr, \qquad \beta \in \widehat{F}. \tag{E.61.1}$$

The pure frame β is unique. Put $h = \beta^{-1}$. There is a unique $\rho \in \widehat{\mathbb{Z}}$ with

$$h^{\mathrm{ab}} = (\rho, \rho), \qquad f(x, y) = h(y, x)^{-1} h(x, y). \tag{E.61.2}$$

Thus ρ is the Nakamura–Schneps frame coordinate in precisely these generators. Define the signed homomorphism

$$q_{12} : \widehat{F} \longrightarrow \mathrm{SL}_2(\mathbb{Z}/12), \qquad x \longmapsto \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad y \longmapsto \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}. \tag{E.61.3}$$

Then every such pair satisfies the exact finite identity

$$q_{12}(f) = \begin{pmatrix} 1 & -8\rho \\ -8\rho & 1 + 64\rho^2 \end{pmatrix} \pmod{12}. \tag{E.61.4}$$

Here is its direct genus-one comparison. In $B_4 = \langle a, b, c \rangle$, put

$$E_1 = (aba)^4, \qquad E_3 = (cbc)^4, \qquad E_2 = cbab^{-1}c^{-1}. \tag{E.61.5}$$

The multiplier-one sewing word, with every frame term retained, is

$$\begin{aligned} \mathcal{R}_\rho(f) = {} & f(E_3, a) c^{-8\rho} f(b^2, c^2) f(E_2, E_3) f(E_1, E_2) \\ & \cdot f(a^2, b^2) a^{8\rho} f(c, E_1) \quad \text{in } \widehat{B}_4. \end{aligned} \tag{E.61.6}$$

For each integer $m \geq 1$, the following is an actual continuous homomorphism, where $(v, M)(w, N) = (v + Mw, MN)$:

$$\begin{gathered} \eta_m : \widehat{B}_4 \longrightarrow (\mathbb{Z}/m)^2 \rtimes \mathrm{SL}_2(\mathbb{Z}/m), \\ a \longmapsto (0, U), \qquad b \longmapsto (0, L), \qquad c \longmapsto (e_1, U), \qquad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}. \end{gathered} \tag{E.61.7}$$

Write $q_m(f) = f(U^2, L^2) = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. Then

$$\eta_m(\mathcal{R}_\rho(f)) = \left(\begin{pmatrix} -B - 8\rho A \\ A - 1 \end{pmatrix}, I \right). \tag{E.61.8}$$

In particular $\eta_{12}(\mathcal{R}_\rho(f)) = 1$ for every normalized full gentle pair of multiplier one. The identity pair and the extra CM pair of E.58 both have this value. Thus this specified sewing projection supplies no additional CM parity restriction.

Proof. We first identify the frame and prove the finite matrix assertion; we then compute the actual four-strand word.

Conjugation by s interchanges x and y. The two expressions for $\Phi(s)$ in (E.61.1) therefore give $f = \beta(y, x)\beta(x, y)^{-1}$. The centralizer calculation in E.58 gives $C_{\widehat{B}}(s) \cap \widehat{F} = 1$, so the pure frame is unique. Write $\beta^{\mathrm{ab}} = (u, v)$. Since $f \in \widehat{F}'$, its abelianization $(v - u, u - v)$ is zero. Hence $u = v$ and $\rho = -u$ proves (E.61.2). This is the word and abelianization used in

[1, Definition 5.1]; it is an identity of the actual frames, with no comparison to a differently normalized Kummer extension.

The image of F modulo four under (E.61.3) is the abelian group

$$V_4 = \left\{ I, \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\}. \tag{E.61.9}$$

Thus $q_4(f) = I$. To keep the sign at three, use the literal matrices

$$S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}. \tag{E.61.10}$$

They define $\widehat{B} \to \mathrm{SL}_2(\mathbb{Z}/12)/\{\pm I\}$, and their restriction to F is the projectivization of (E.61.3). Indeed $SR = U^{-1}$, $x = (sr)^{-2}$, and $y = sxs$. The two possibilities for the global sign are distinguished modulo four, since $-I \notin V_4$.

Put $g = q_{12}(\beta)$. The signed identity

$$q_{12}(f) = SgS^{-1}g^{-1} \tag{E.61.11}$$

follows from the free-generator substitution, because S conjugates U^2 to L^2 and conversely. Since $\Phi(r)^3 = 1$, the matrix $(q_{12}(f)^{-1}R)^3$ is I or $-I$. Its reduction modulo four is $R^3 = I$, so it is I. This proves the signed cubic condition at three, without discarding the central sign.

For completeness, the required order-twenty-four calculation has the following exact form. Work in $\mathrm{SL}_2(\mathbb{F}_3)$, and put $X = U^2$. The character

$$\epsilon : \mathrm{SL}_2(\mathbb{F}_3) \longrightarrow \mathbb{F}_3, \qquad \epsilon(X) = \epsilon(L^2) = 1 \tag{E.61.12}$$

has quaternion kernel. Its existence and the complete enumeration can also be checked by the decomposition $\mathrm{SL}_2(\mathbb{F}_3) = \langle X \rangle Q_8$, where $Q_8 = \langle S, \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \rangle$. For g in this group,

$$\left((SgS^{-1}g^{-1})^{-1}R\right)^3 = I \quad \Longleftrightarrow \quad g = X^k S^j \quad (0 \le k < 3,\ 0 \le j < 4). \tag{E.61.13}$$

The remaining twelve matrices give $-I$ for the cube. Direct multiplication on the twelve displayed matrices gives

$$SgS^{-1}g^{-1} = \begin{pmatrix} 1 & k \\ k & 1 + k^2 \end{pmatrix}, \qquad \epsilon(g) = k. \tag{E.61.14}$$

These identities check every element, not merely the saved CM representatives. Now $\beta^{\mathrm{ab}} = (-\rho, -\rho)$, so $\epsilon(g) = -2\rho = \rho$ modulo three. Equations (E.61.11)–(E.61.14) prove (E.61.4) modulo three. Equation (E.61.9) proves it modulo four, and the Chinese remainder isomorphism proves the signed modulus-twelve assertion.

The three matrices in (E.61.7) satisfy the two adjacent braid relations and the far commutation relation. They therefore give the indicated homomorphism. Its linear part is the literal Ferrari fold $B_4 \to B_3$, $a, c \mapsto a$, $b \mapsto b$, followed by the standard integral matrix representation. The Ferrari generators ca^{-1} and $abac^{-1}b^{-1}a^{-1}$ map respectively to e_1, e_2. Thus the translation module and its basis are fixed by actual words.

Both E_1 and E_3 map to the identity under η_m. For example the linear part of cbc is ULU, whose square is $-I$, and its four-term translation sum is zero. Since $f \in \widehat{F}'$, all four factors in (E.61.6) involving E_1 or E_3 therefore disappear. We are left with

$$\eta_m(\mathcal{R}_\rho(f)) = \eta_m(c)^{-8\rho} f(\eta_m(b)^2, \eta_m(c)^2)(0, q_m(f))\eta_m(a)^{8\rho}. \tag{E.61.15}$$

Put $w = (0, -1)^t$ and $t_w = (w, I)$. The two exact identities

$$t_w(0, L^2)t_w^{-1} = (0, L^2), \qquad t_w(0, U^2)t_w^{-1} = (2e_1, U^2) = \eta_m(c)^2 \tag{E.61.16}$$

prove, by the universal property of $\widehat{F}$, that

$$f(\eta_m(b)^2, \eta_m(c)^2) = \big((I - q_m(f)^{-1})w, q_m(f)^{-1}\big). \tag{E.61.17}$$

Here $f(L^2, U^2) = q_m(f)^{-1}$ is precisely the gentle symmetry identity. The vector in (E.61.17) is $(-B, A - 1)^t$. Multiplying the four factors of (E.61.15) now gives (E.61.8), including its signs and the factor eight.

We record the exact sewing-source comparison to prevent the discarding of a central braid factor. Write

$$X_{12} = a^2, \quad X_{23} = b^2, \quad X_{34} = c^2, \quad X_{45} = (aba)^2, \quad X_{51} = (cbc)^2, \quad Z = (abc)^4. \tag{E.61.18}$$

Let $A_3 = \langle u, v \mid uvuv = vuvu \rangle$ be the index-three subgroup $\langle \tau_1, \tau_2^2 \rangle$ of B_3. The four actual homomorphisms to B_4 used by the sewing calculation send (u, v) to

$$(a, X_{51}), \quad (E_2, X_{51}), \quad (E_2, X_{45}), \quad (c, X_{45}). \tag{E.61.19}$$

Their central squares $(uv)^2$ are, in order,

$$X_{34}Z, \quad X_{23}Z, \quad X_{23}Z, \quad X_{12}Z. \tag{E.61.20}$$

These are equalities in the discrete braid group. The certificate checks the defining relation and each square through the faithful Artin representation, whose generators are written explicitly in the output; faithfulness in this convention is [3, Theorem A, representation (1) with $k = 1$]. Thus the maps extend to profinite completions.

For a pair in the Nakamura–Schneps subgroup, relation (IV) of [1, Section 1] reads

$$f(u, v^2) = v^{4\rho} f(u^2, v) u^{4\rho} (uv)^{-4\rho}. \tag{E.61.21}$$

Applying the four maps, with inversion on the first and third expressions, retains the following full braid formulas:

$$\begin{aligned} f(E_3, a) &= Z^{2\rho} X_{34}^{2\rho} X_{12}^{-2\rho} f(X_{51}, X_{12}) X_{51}^{-4\rho}, \\ f(E_2, E_3) &= Z^{-2\rho} X_{51}^{4\rho} f(X_{14}, X_{51}) X_{14}^{2\rho} X_{23}^{-2\rho}, \\ f(E_1, E_2) &= Z^{2\rho} X_{23}^{2\rho} X_{14}^{-2\rho} f(X_{45}, X_{14}) X_{45}^{-4\rho}, \\ f(c, E_1) &= Z^{-2\rho} X_{45}^{4\rho} f(X_{34}, X_{45}) X_{34}^{2\rho} X_{12}^{-2\rho}. \end{aligned} \tag{E.61.22}$$

Here $X_{14} = E_2^2$. The central powers in their product cancel with coefficients $2, -2, 2, -2$. This is the literal multiplier-one word of [1, proof of Claim 8.3, relation (R')], which is also the genus-one sewing calculation invoked in [2, Theorems 3.18 and 4.9]. Equation (E.61.22) is the source's calculation on its Nakamura–Schneps subgroup; the proof of (E.61.4) and (E.61.8) did not impose (IV) on a full-GT element.

The program `bc_genus_one_sewing.py` verifies all sixteen displayed braid comparisons, the complete twenty-four-element matrix classification, the translation conjugator, and the actual framed words of E.1. Its JSON retains the four-generator Artin images, the twelve passing and twelve failing cubic matrices, the exact recovered β words and ρ values, and every mod-three, mod-four and mod-twelve sewing value. All twenty-four saved CM words give the identity in these projections. The initial search for a change under the $4,656$ literal N'_B commutator corrections to β found none; it is preserved under `research/E61_initial_affine_probe`. The universal calculation above explains that outcome. □

References

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